

1.7. Using define() and require() in RequireJS to Create and Use JavaScript Modules

Purpose:

To understand how RequireJS helps modularize JavaScript code by defining reusable logic in one file (define()) and consuming it in another (require()).

Pre-requisites:

- Basic JavaScript knowledge (functions and variables)
- Familiarity with running HTML/JS in a browser
- RequireJS script linked in your HTML

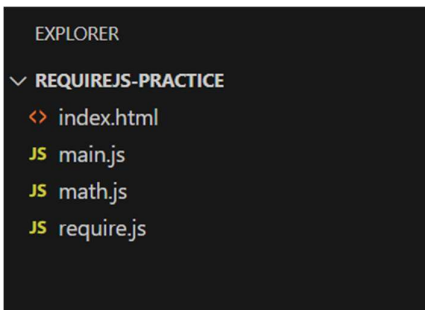
Concepts Covered:

- AMD (Asynchronous Module Definition)
- define() to create a module
- require() to use one or more modules
- Benefits of modular code
- Basic example with math operations

Steps to Perform:

1. Create a folder requirejs-demo and inside it, create these files, use VS code:

- math.js
- main.js
- index.html
- require.js (downloaded from <https://requirejs.org/docs/download.html> or via CDN)



2. For RequireJs file : you need to

- **Download** RequireJS from here

<https://requirejs.org/docs/release/2.3.6/minified/require.js> (it's a minified version of requireJs)



```

/** vim: et:ts=4:sw=4:sts=4
 * @license RequireJS 2.3.6 Copyright jQuery Foundation and other contributors.
 * Released under MIT license, https://github.com/requirejs/requirejs/blob/master/LICENSE
 */
var requirejs,require,define;!function(global,setTimeout){var req,s,head,baseElement,dataMain,src,interactiveScript,cb
([!:"="]|^)\./.*$/gm,cjsRequireRegExp=/^[^.]?require\s*\(\s*['"]([^\s"]+)[\s"]\s*\)/g,jsSuffixRegExp=/\.js$/,currDir
("undefined"==typeof window||"undefined"==typeof navigator||!window.document),isWebWorker=!isBrowser&&"undefined"!="ty
3"==navigator.platform?/^complete$/:^(complete|loaded)$/,defContextName=".",isOpera="undefined"!="typeof opera&&"[ob
],useInteractive=!1;function commentReplace(e,t){return t||""}function isFunction(e){return"[object Function]"==ost
{var i;if(e)for(i=0;i<e.length&&!e[i]||!t(e[i],i,e));i+=1);function eachReverse(e,t){var i;if(e)for(i=e.length-1;-1
getOwn(e,t){return hasProp(e,t)&&e[t]}function eachProp(e,t){var i;for(i in e)if(hasProp(e,i)&&t(e[i],i))break}functi
e||e||isArray(e)||isFunction(e)||e instanceof RegExp?i[t]=e:[i[t]]=[i[t]||{}],mixin(i[t],e,r,n))}},i}function bind(e
document.getElementsByTagName("script"))function defaultOnError(e){throw e}function getGlobal(e){if(!e)return e;var t
Error(t+"\\nhttps://requirejs.org/docs/errors.html#"+e);return n.requireType=e,n.requireModules=r,i&&(n.originalError=
{if(isFunction(requirejs))return;cfg=requirejs,requirejs=void 0}void 0===require||isFunction(require)||{cfg=require,r
isArray(e)||"string"==typeof e||(o=e,isArray(t)?(e=t,t=i,i=r):e=[]),o&&o.context&&(a=o.context),(n=getOwn(contexts,a)
(n=contexts[a]=req.s.newContext(a)),o&&n.configure(o),n.require(e,t,i)},req.config=function(e){return req(e)},req.nex
(require=req),req.version=version,req.jsExtRegExp=/^\/\?:\?\.js$/g,req.isBrowser=isBrowser,s=req.s={contexts:contexts
ren[t]=function(){var e=contexts[defContextName];return e.require[t].apply(e,arguments)}}}.isBrowser&&(head=s.head&&

```

- Save it as **require.js** in your project folder . Mine is “**requireJS-practice**”.

3. Then create **math.js** file in same folder and write the following code

```

define([], function () {
    return {
        add: function (a, b) {
            return a + b;
        },
        subtract: function (a, b) {
            return a - b;
        }
    };
});

```

Explanation:

- This file defines a **reusable module** named math.
- It uses define() — part of the AMD (Asynchronous Module Definition) pattern.
- define([], function () { ... }) tells RequireJS:
 - No other modules are needed here ([] means no dependencies).
 - The function returns an **object with two methods**: add and multiply.
- This module can now be reused anywhere by simply requiring 'math'.

4. Then create **main.js** file in same folder and write the following code

```
require(['math'], function (math) {
  console.log("Add: ", math.add(5, 3));           // Outputs: Add: 8
  console.log("Subtract: ", math.subtract(5, 3)); // Outputs: Subtract: 2
});
```

Explanation:

- This is the **entry point JavaScript file** for our app.
- It tells RequireJS: "I want to use the math.js module."
- The require() function loads the math module **asynchronously**.
- Once loaded, the math module is available inside the callback function.

Because modules in RequireJS load asynchronously, we use a callback function to make sure our code only runs after the 'math' module is fully loaded.

- Inside the callback, we call the add() and multiply() methods from the math module and log the results.

5. Then create **index.js** file in same folder and write the following code

```
<!DOCTYPE html>
<html>
<head>
  <title>RequireJS Example</title>
  <script data-main="main" src="require.js"></script>
</head>
<body>
  <h2>Open the console to see math module output</h2>
</body>
</html>
```

Explanation:

- This is the **main HTML file** that launches everything.
- <script> tag loads **RequireJS** using the src.
- “<script data-main="main" src="require.js"></script>”

It means:

- ❖ "Load the file named main.js as the starting point for my application."

- ❖ **src="require.js"** → This loads the RequireJS library.
- ❖ **data-main="main"** → Tells RequireJS:
- ❖ After loading itself,
- ❖ **It should automatically load a JavaScript file called main.js.**
- ❖ **It assumes the file is named main.js, so you don't have to write .js.**
- When the browser opens this page, RequireJS will load main.js, which in turn may load other files (modules).

6. Output:

Open the browser → Right-click → Inspect → Console tab:

You'll see:

